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In silico analysis of novel Triacontafluoropentadec-1-ene as a sustainable replacement for dodecane in fisheries microplastics: Molecular docking, dynamics simulation and pharmacophore studies of acetylcholinesterase activity

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ABSTRACT

Plastics play an essential role in modern fisheries and their degradation releases micro- and nano-sized plastic particles which further causes ecological and human health hazards through various environmental contamination pathways and toxicity mechanisms, which can cause respiratory problems, cancer, reproductive toxicity, endocrine disruption and neurological effects in humans. This study utilized various bioinformatics tools through multi-step computational analyses to investigate the interactions between prevalent fisheries microplastics and the key protein receptor acetylcholinesterase (AChE), which is associated with neurotoxicity, as it can interfere with nerve impulses and muscle control. Our results indicate that the binding of seven polymers within AChE's active site, with dodecane and polypropylene exhibited highest affinity with hydrogen bonding were observed through Molecular docking of different program (PyRx) and servers (CB-Dock, eDock) then the stability of AChE-dodecane and AChE-polypropylene complexes were observed through MD simulations for 100 ns. Further analysis of dodecane was done by using pharmacophore modelling and virtual screening. The pharmacophore model of dodecane is based on six hydrophobic rings. Using this model, we screened among thousands of substrates form (CMNPD, COCONUT, NPASS, NANPDB, and ZINC) database and identified fifty highly similar candidates that align with dodecane's structure and interaction with acetylcholinesterase (AChE). The compound triacontafluoropentadec-1-ene exhibited highest binding affinity (score: -9.6) which was further confirmed through molecular dynamics for 100 ns. The key finding for this study is triacontafluoropentadec-1-ene as a promising alternative to dodecane, and the study highlights that the integrated *in silico* framework presents a valuable computational model for guiding future guidelines on environmental safety through prioritizing constituents and accelerated discovery of alternatives. These findings will help us identify the most hazardous plastics through ranking and characterizing the substance for sustainably "greening" fisheries worldwide. The study forecasts the groundwork of these compounds, which may be able to reduce the environmental toxicity of microplastics in future.

1. Introduction

The global plastic production has increased from 1.5 million tons in 1950–359 million tons in 2018 and A significant portion of these plastics, estimated between 4.8 and 12.7 million tons annually, reaches the ocean through various pathways. Plastics have revolutionized fishery industries due to their flexibility and durability but have also caused environmental problems worldwide (Jenkins et al., 2022; Veerasingam et al., 2020). According to the annual report of Central Pollution Control Board (CPCB) for the year 2018–19, the total plastic waste generation in India was approximately 9.46 million tonnes per year. The major contributors were Tamil Nadu (12 %), Maharashtra (12 %), Gujarat (11 %), West Bengal (9 %), Uttar Pradesh (8 %), Karnataka (8 %), and Delhi (7 %) (Rafey and Siddiqui, 2023). Plastics commonly used in fisheries include polyethylene (PE), polyamide-66 (Nylon-66), polycarbonate (PC), polyethylene terephthalate (PET), polyurethane (PU), polypropylene (PP) and polyvinyl chloride (PVC) (Gligorijevic et al., 2024; Issac and Kandasubramanian, 2021). Long-term uses of plastic in all these processes caused leaching and "weathering" leading to microplastics (MP) and nanoplastic (NP) formation (Liu et al., 2020b). Among these plastics, PE plays a pivotal role in supporting the global fishing industry. Overall, the hydrocarbon dodecane provides the raw material to produce versatile PE polymers that are essential across applications sustaining fisheries worldwide (Ishihara et al., 2014). Lightweight and durable PE threads are chiefly used in fishing nets since they offer strength without losing buoyancy in water. Long-term usage and weathering of PE results in dodecane release in marine environment (Petrulionien˙e et al., 2023). It is estimated that each year between 1.15 and 2.41 million tons of plastic flow from the global riverine system into the ocean from various source of larger plastic waste, the shedding of synthetic textiles, the wear and tear of automobile tires, and the breakdown of personal care and household products (Carney Almroth et al., 2018; Kole et al., 2017; Lebreton et al., 2017), and those plastic materials generating 'microplastics' ($\leq$1 mm in diameter) and 'nanoplastics' ($\leq$1 μm in diameter) create problems for marine life and environments (Gigault et al., 2021). Generally with lifespans of two to five years, nylon and polyester nets, lines and cages gradually release microfibers during degradation, causing microplastics and eventually NP pollution (Avinash et al., 2023; Erasmus et al., 2024). Studies has shown that certain commercial fish species and seafood contain NP, which raises concerns related to potential toxicity and bioaccumulation (Bhagat et al., 2022). MP, due to their microscopic size, can plausibly be mistaken as prey during regular feeding. Intake of MP induces ulcer, erosion and lesion in the fish gut (Bhagat et al., 2022). Excessive microplastics accumulations might fully clog intestines and toxins leaching from MP also introduce risks to health (Wang, 2023). Additionally, MP could transport persistent pollutants such as, Dichlorodiphenyltrichloroethane (DDT), Dodecane, Polychlorinated Biphenyls (PCBs) and Polycyclic Aromatic Hydrocarbons (PAHs) into fish tissues via bioaccumulation (Baqar et al., 2018), increasing threats for higher trophic levels including humans. It can cause respiratory problems, cancer, reproductive toxicity, endocrine disruption and neurological effects in humans.

Acetylcholinesterase (AChE) is a neurotransmitter. It has an essential function in terminating neurotransmission at cholinergic synapses as a serine hydrolase with a catalytic triad breaking down acetylcholine (Dvir et al., 2010). In vertebrates like fish and humans, AChE is essential for synaptic transmission in the neuromuscular junction, autonomic ganglia, central nervous system, and neuroendocrine tissues (Brandt and Flurie, 2020). Also, AChE is an important enzyme involved in the termination of nerve impulse transmission at cholinergic synapses by rapidly hydrolysing the neurotransmitter acetylcholine (Nachon et al., 2013; Tripathi and Srivastava, 2008). Specifically in fish, AChE acts throughout the brain, spinal cord, nerves, skeletal muscles and heart (Kroth et al., 2017). Given its conserved structure and important enzymatic role, AChE serves as a key molecular target associated with disruptions in neuromuscular, autonomous and central nervous system (CNS) functions in both fish and humans populations (Brandt and Flurie, 2020). Exploring interactions between AChE and molecular components of MP could deliver insights into mechanisms of neurotoxicity with broad ecological and public health consequences.

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Although there is ongoing research on MP in aquatic systems, knowledge gaps remain regarding fisheries-derived sources, properties, and impacts, particularly regarding nano-sized plastics, which may be more toxic due to cellular absorption. Little data exists on nano-plastic pollution introduced through fishing operations (Namasivayam and Avinash, 2024). To address this, the study aims to utilize computational methods such as molecular docking, pharmacophore modelling, and MD simulation to predict the molecular level interactions between diverse plastics and biological receptors (AChE), providing insights into their toxicity mechanisms (Chagaleti et al., 2024; Rana and Sivaperumal, 2018). This will help us identifying the most hazardous plastics through ranking. The goal of this study aims to characterize the impacts of under-investigated nano-sized and fisheries-specific plastic pollution properties, arranging the substance for sustainably "greening" fisheries worldwide while considering environmental, economic and social outcomes.

2. Materials and methods

As a first step, the physicochemical properties and toxicity data of potential plastic components were collected from databases to identify MP components. Next, Absorption, Distribution, Metabolism, Excretion and Toxicity (ADMET) analysis was performed in-silico using admetSAR2 and admetLab servers to predict the properties of the MP (Xiong et al., 2021; Yang et al., 2019). Seven microplastics were initially evaluated and selected based on the literature review and ADMET profiling results. Then, the crystal structure of the AChE receptor (4BDT) was retrieved from the PDB database (Nachon et al., 2013). This receptor structure was prepared for molecular docking simulations. Three-dimensional structures of the selected plastic ligands were obtained from PubChem. Molecular docking simulations using PyRx, CB-Dock and eDock programs were conducted to predict how the microplastics might bind to the AChE receptor. Selected docking poses underwent molecular dynamics simulations with GROMACS to study the ligand-receptor interactions over time (Packiapalavesam et al., 2024). Finally, a pharmacophore model was generated based on the docking results (Pal et al., 2019). This pharmacophore was then used to virtually screen databases to identify substitute microplastics ligands that could bind the receptor. The new ligands identified through virtual screening also underwent ADMET, docking and molecular dynamics simulations and analysis.

2.1 ADMET analysis

The admetSAR2 (http://lmmd.ecust.edu.cn/admetsar2) server and admetLab (https://admetlab3.scbdd.com/server/screening) platform were used to estimate Absorption, Distribution, Metabolism, Excretion and Toxicity (ADMET) properties of microplastics (Xiong et al., 2021; Yang et al., 2019). ADMET profiles generated by admetSAR2 using machine learning models and data-driven algorithms provided physicochemical properties such as solubility and permeability contributing to absorption. Toxicity risks covering carcinogenicity, mutagenicity and reproductive/developmental effects were estimated. AdmetLab complemented admetSAR2 by providing additional ADMET modules to assess microplastics, computing quantitative toxicity endpoints including acute toxicity in rats and chronic human health hazards quantified as LD50, NOAEL or IC50 values (Đuki´c-´Cosi´c et al., 2021).

2.2 Ligands retrieval and preparation

Seven plastic components commonly used in the fishing industry were retrieved from the PubChem database (https://pubchem.ncbi.nlm.nih.gov/). These included polyamide-66 (Nylon-66, CID=36070), polyvinyl chloride (PVC, CID=6338), polycarbonate (PC, CID=6623), polypropylene (PP, CID=7923), polyethylene terephthalate (PET, CID=18721140), polyurethane (PU, CID=12254) and dodecane (CID=8182) (Table 1). The SDF files contained detailed stereochemical

Table 1 *Plastic compounds and their PubChem CID. List of microplastics chemical components which are used for protein-ligand interaction analysis.*

Compound	PubChem CID
Polyamide (nylon–66)	36,070
Polyvinyl chloride	6338
Polycarbonate	6623
Polyethylene terephthalate	18,721,140
Polyurethane	12,254
Polypropylene (PP)	7923
Dodecane	8182

representations of each polymer's characteristic repeating unit. These 3D structures were retrieved for use as candidate ligands in subsequent docking simulations. The 3Dstructure of Plastic components are downloaded from PubChem compound search of NCBI. The structures were downloaded as.sdf file format and open through Discovery Studio v24.1.0.23298 visualization tool and converted into.pdb file.

2.3 Protein structure retrieval and preparation

The crystal structure of human AChE was retrieved from the Protein Data Bank (RCSB PDB: https://www.rcsb.org/). The selected structure was human AChE (pub id: 4BDT) resolved at 3.10 Å with all 583 amino acid residues built into the electron density map. AutoDock Tools v.1.5.7 was used to prepare the 4BDT PDB file for docking. Non-protein entities such as water were deleted and polar hydrogens were added. Kollman united atom forcefield and Gasteiger charges were assigned to generate PDBQT files compatible with AutoDock Vina. Protonation states of the catalytic triad (Ser198, Glu327 and His440) and oxyanion hole (Gly116, Gly118 and Ala201) residues were optimized to represent physiological pH 7.4.

2.4 Binding side prediction

The CASTp server was used for computational analysis of protein structure and function to predict the structure and binding pocket of human AChE (http://sts.bioe.uic.edu/castp/) (Tian et al., 2018). Once, the pdb id is being write down and structural model has been generated, CASTp evaluates cavities in proteins systematically; evaluating them using parameters such as accessible surface area, depth, etc., to identify probable binding sites. In this investigation, the amino acid sequence of AChE was subjected to the CASTp server which predicted the 3D structure shown above that pointed out largest and most suitable pocket for ligand-binding coinciding.

2.5 Molecular interaction with plastic components

2.5.1 Molecular docking using PyRx

To understand the binding interactions of AChE and microplastics, molecular docking were performed using PyRx 0.8 was as a DockingBased Virtual Screening Tool (Dallakyan and Olson, 2015). PyRx performs Blind docking with Lamarckian Genetic Algorithm (LGA) simulations which explores a wide range of conformational space without any bias towards it (Adi et al., 2016). An active site grid box of dimensions 61.4943 × 101.3277 × 70.2490 Å were generated, centered at coordinates 0.4991, -47.4445, -58.7426 Å. In support of the flexible docking calculations for each simulated microplastics upon the AChE binding pocket with AutoDock Vina implemented in PyRx; while the top ten ligand poses were selected based on predicted binding free energy values obtained from this program. Discovery Studio software was used to visually analyzed protein-ligand interactions such as hydrogen bonds, π-π interactions and hydrophobic contacts of binding between microplastics bind to AChE.

2.5.2 Molecular docking using CB-Dock Server

The 3D structure of prepared AChE was used as the receptor and the Seven MP were docked individually using the CB-Dock server (https://cadd.labshare.cn/cb-dock/php/blinddock.php) (Cao and Li, 2014; Liu et al., 2020a). AutoDock Vina, were selected for docking with default parameters (Liu et al., 2020a). CB-Dock performed parallel docking runs of each ligand against the AChE target using the selected programs. The docked poses were clustered and the top ranked clusters were visualized. Protein-ligand interactions such as hydrogen bonds, π-stacking and hydrophobic contacts were analysed using the 3D viewer. Binding free energies were predicted using the scoring functions of the respective docking programs. This allowed evaluation and comparison of the binding modes predicted by different tools on the CB-Dock server.

2.5.3 Molecular docking using eDock server

The Seven MP were used as ligands while the 3D structure of prepared AChE used as a receptor. The prepared AChE.pdb file were uploaded to the eDock server (https://zhanggroup.org/EDock/) with default parameters (Zhang et al., 2020). Docking runs were performed using eDock's molecular mechanics force field. The discovery studio visualizer used for analysing protein-ligand interactions such as π-π stacking, hydrophobic contacts, and hydrogen bonds among others on the top predicted poses. Favourable binding modes between structural molecules of microplastics and structural molecules of AChE were calculated by the binding energies based on eDock DSS scoring function (Kataria et al., 2024).

2.6 Molecular docking with AChE substrate

The three-dimensional crystal structure of acetylcholinesterase protein bound with huprine W was retrieved from the Protein Data Bank in PDB format (pdb id: 4BDT). The sdf file of the substrate huprine W (C18H19ClN2O) was downloaded from the PDB database (Nachon et al., 2013). Huprine W was prepared using the PyRx tool by generating conformers and assigning Gasteiger charges. Molecular docking of huprine W with the prepared AChE protein was performed using PyRx.

2.7 Pharmacophore modeling, molecular interaction and ADMET analysis

The AchE receptor structure was uploaded to the Pharmit platform (http://pharmit.csb.pitt.edu) under the "Receptors" section to define the target binding site constraints (Sunseri and Koes, 2016). The generated Dodecane pharmacophore model was also loaded under the "Pharmacophore" module to enable pharmacophore-based searches of ligand databases (Rana and Sivaperumal, 2018).

2.7.1 Molecular interaction studies

The fifty screened hits from the previous pharmacophore modelling and filtering were subjected to molecular docking simulations against the prepared AChE receptor using the PyRx software. The active site grid encompassing key residues was generated based on the co-crystallized dodecane position. The energy minimization of each ligand was carried out by PyRx to allow flexible rotations during blind docking employing Lamarckian Genetic Algorithm for conformational searching without bias (Adi et al., 2016). Ten independent docking runs were performed for every ligand accounting for potential local minima (Mansuri et al., 2016). The resulting poses were clustered, binding affinities ranked and favorable interactions identified using PyMOL to gain qualitative insights into hit binding modes and recognize substitutes resembling endogenous substrate binding for further evaluation.

2.7.2 ADMET analysis

The AdmetSAR2 server and AdmetLab servers were used to estimate Absorption, Distribution, Metabolism, Excretion and Toxicity (ADMET) properties of Triacontafluoropentadec-1-ene.

2.8 Molecular dynamics simulation of protein and ligand complexes

The AChE-Dodecane, AChE-Polypropylene and AChE-Triacontafluoropentadec-1-ene complex structure was prepared using UCSF Chimera(Pettersen et al., 2004). Hydrogen atoms were added to the AChE protein and ligand using Chimera's Probe function. The ligand was parameterized using SwissParam to generate CHARMM36 forcefield parameters and topology files (Zoete et al., 2011). The prepared receptor and ligand files were saved in PDBQT format and collected in a GROMACS workspace (Abraham et al., 2015). The ligand coordinates were manually copied from LIG.mol2 to LIG.gro to maintain the docked pose relative to the protein in REC.pdb. The topology file was compiled from the forcefield parameters and atom names of receptor and ligand files. The complex was solvated using gmx solvate and equilibrium simulated with two phases - NVT at 300 K with position restraints followed by 100 ps NPT at 1 bar. A 100 ns production molecular dynamics simulation was performed with leapfrog integration, LINCS constraints and periodic boundary conditions to sample the solvated complex (Berendsen et al., 1995; Van Der Spoel et al., 2005).

3. Results and discussion

The first step in the computational analysis was to retrieve the 3D structure of acetylcholinesterase, which is a key target for microplastics toxicity. The crystal structure of AChE from *Homo sapiens* (Protein Data Bank ID: 4BDT) 3D structure of AChE.

3.1 ADMET (Absorption, distribution, metabolism, excretion, and toxicity) analysis

The ADMET properties describe aspects of absorption, distribution, metabolism, excretion and toxicity of chemicals. Knowledge of the ADMET characteristics of plastic polymers is crucial for understanding their fate within biological systems and potential hazardous effects.

3.1.1 Admetsar2 ADMET result

The in-silico ADMET predictions from admetSAR2 provided useful understanding of the possibilities in absorption, distribution, metabolism and toxicity profiles of the seven plastic polymers. Intestinal absorption was predicted to be high (>0.9) for all chemicals except PET, which showed moderately high absorption of + 0.9681. Caco-2 cell permeability as a measure of epithelial transport varied more widely, from -0.6045 for PP to + 0.9769 for dodecane, indicating potential differences in transcellular absorption. All chemicals except polycarbonate (+0.5750) and polyamide (+0.9750) were predicted as nonpermeant to the blood brain barrier, meaning the latter two may more readily enter the central nervous system after systemic exposure and warrant further examination of any neurotoxic effects. Oral bioavailability ranged from -0.6429 to + 0.8286, with moderately high scores predicted for dodecane, PP and PVC but the lowest for polyamide (Supplementary Table. 1). Variability in subcellular localization extended from 0.4945 to 0.8727, with polycarbonate estimated to accumulate most readily in certain organelles. Membrane transporter interactions were also predicted to vary widely. Toxicity risk scores were obtained for endpoints like carcinogenicity, genotoxicity, local irritancy and specific organ toxicities, with some hazard flags for plastics such as PVC and dodecane. Fish and crustacean aquatic toxicity predictions revealed a broad range of values, from -0.9166 to + 0.9770 for fish and -0.5400 to + 0.8690 for crustaceans. Dodecane consistently exhibited the highest hazard scores for both, likely due to its high logP of 3.82 favouring accumulation over more water-soluble plastics such as polycarbonate and polyamide based on estimated solubility ranging from -1.497 mg/L for nylon-66 to -5.178 mg/L for dodecane (Supplementary Table. 1).

3.1.2 AdmetLab ADMET result

The AdmetLab ADMET results show that Polycarbonate, polyethylene terephthalate, polyurethane, PP and polyamide showed high predicted gastrointestinal absorption, indicating oral exposure could lead to substantial absorption from contaminated food/water. In contrast, PVC showed only low absorption. Dodecane showed the h ighest predicted skin permeation (log Kp -3.01 cm/s), while PU and PP were the least permeable (-7.36 and -6.99 cm/s) (Supplementary Table. 2). PC,

PET and polyamide were predicted to cross the blood-brain barrier, raising neurological concerns following oral/inhalation uptake, while the others were not predicted to permeate the blood-brain barrier. None were predicted to be P-glycoprotein substrates. Polycarbonate was predicted to inhibit CYP1A2 and CYP2D6, while dodecane may inhibit CYP1A2. Molecular weight, hydrogen bond acceptors/ donors and calculated logP values provided context for solubility, permeability and partitioning tendencies. For example, polycarbonate had the highest molecular weight (228.29 g/mol), lowest water solubility but greatest predicted lipophilicity, indicating biomagnification potential, while PVC had the lowest molecular weight (62.5 g/mol), lowest lipophilicity and greatest aqueous solubility, predicting lesser bioaccumulation risk (Supplementary Table. 2). Overall, dodecane and polycarbonate displayed the most concerning ADMET properties based on high absorption, blood-brain barrier permeation, CYP inhibition and lipophilicity.

The ADMET analyses conducted using AdmetSAR2 and AdmetLab tools provided largely consistent and useful insights into the comparative absorption, distribution, metabolism and toxicity properties of the seven common plastic polymers (Xiong et al., 2021; Yang et al., 2019). Both tools predicted high intestinal absorption for most plastics except PVC, with dodecane showing the greatest absorption potential due to its high logP value. Specifically, the tools identified polycarbonate and dodecane as posing particular ADMET hazards, with polycarbonate predicted to cross the blood-brain barrier and inhibit cytochrome P450 enzymes, while dodecane was calculated to have very low aqueous solubility and highest bioaccumulation risk. This in-silico ADMET analysis was thus able to validate the relative toxicities of the plastic components (Uzzaman et al., 2021), revealing polycarbonate and dodecane (polyethylene) as highly toxic, with PP posing the lowest toxicity risk.

3.2 Binding side prediction and molecular interaction analysis

The computational prediction of the binding pocket on the AChE protein structure identified five potential sites. The highest ranked binding pocket, designated as site 1, has an area of 383.297 square angstroms and a volume of 175.302 cubic angstroms. The second highest ranked site, site 2, has dimensions of 159.063 square angstroms for its area and 151.624 cubic angstroms for its volume. The third site has an area of 119.602 square angstroms and a volume of 89.972 cubic angstroms. Site 4 has an area of 145.592 square angstroms and a volume of 72.363 cubic angstroms. The fifth site has a predicted area of 112.760 square angstroms and a volume of 53.976 cubic angstroms data. The appended image demonstrates localization of the highest ranked binding pocket, site 1, on the three-dimensional protein structure depicted in grey. The red highlighted region corresponds to this predicted binding pocket. These computational results provide a framework for experimentally probing ligand interactions within the top ranked pockets identified on the AChE protein surface.

3.2.1 Molecular docking using PyRx tool

The molecular docking simulations provided insights into how the various plastic polymers and dodecane interact within the active site of AChE. Dodecane (-5.1 kcal/mol) showed the second highest affinity, primarily through hydrophobic π-σ interactions between the carbons of its alkyl chain and the aromatic side chains of Tyr341 and Trp286 located inside the catalytic pocket. These hydrophobic interactions serve to sequester dodecane close to the catalytic triad, while an additional π-alkyl bond with Tyr124 provides further binding to the active site. This strongly lipophilic binding of dodecane is believed to cause steric blockage through van der Waals entrapment rather than polar bonding. Polycarbonate exhibited the greatest binding affinity of -8.2 kcal/mol, attributable to its interaction profile involving multiple hydrogen bonds donated by ester groups to Tyr124, Tyr72, and Tyr337 anchoring one end of the polymer (Table 2). Additionally, one of its aromatic ring's forms π-π stacking with the indole ring of Trp286 located at the peripheral anionic site near the entrance. Hydrogen bonding and π-donor interactions fix polycarbonate into a bridging configuration interacting jointly at both ends of the active site cleft. PU engaged in various hydrogen bonding interactions from Trp236 and Phe297 near the base to Val239 in the AChE subsite as well as van der Waals contacts, connecting significant lengths of the cleft through polar and nonpolar interactions to form an interaction barrier, albeit weaker than polycarbonate. Similarly, PP utilized hydrogen bonds donated along the active site from Lys53 to Asn186 and Gly14 as well as alkyl and π-π interactions up to Trp182 (Table 2). Interestingly, the lowest affinity binder PVC engaged residues through specially chosen nonpolar alkyl interactions with chlorinated carbons despite fewer overall interactions. Polyamide application of van der Waals/π-alkyl contacts at the entrance and hydrogen bonding in the substrate enables inhibitory obstruction of substrate passage.

Table 2 *This table summarizes the plastic components docked against the AChE protein receptor using three different molecular docking tools: PyRx, CB-Dock, and eDock. For each tool, the compound with the highest binding affinity (kcal/mol for PyRx) and score (Vina score for CB-Dock and X-Score for eDock) was selected for further analysis.*

Plastic components	PyRx binding affinity (kcal/mol)	CB-Dock Vina score	eDock X-Score	Bonds and interactions
Polyurethane	−3.9	−4.2	4.343	**PyRx:** Conventional Hydrogen Bond (TRP A:236, PHE A:297, ARG A:247, PRO A:235) and Alkyl(VAL A:239). **CB-Dock:** Conventional Hydrogen Bond (PHE A:297, TRP A:236, ARG A:247, PRO A:235). **eDock:** Conventional Hydrogen Bond (TYR A:337, SER A:203) and Pi-Alkyl(HIS A:447).
Dodecane	−5.1	−5.8	5.310	**PyRx:** Pi-Sigma (TYR A:341, TRP A:286) and Pi-Alkyl(TYR A:124). **CB-Dock:** Pi-Sigma(TRP A:439) and Pi-Alkyl(PHE A:297, TRP A:86, TYR A:124, TYR A:337). **eDock:** Alkyl(VAL A:294) and Pi-Alkyl(HIS A:447, PHE A:338, TYR A:124, PHE A:297, TYR A:341).
Polypropylene	−4.7	−4.7	4.450	**PyRx:** Conventional Hydrogen Bond (LYS A:53, ASN A:186, GLY A:14) and Pi-Pi T-shaped(TRP A:182) and Pi-Alkyl(LYS A:53). **CB-Dock:** Conventional Hydrogen Bond (TYR A:337, PHE A:295, TYR A:124) and Pi-Pi Stacked(TYR A:341). **eDock:** Conventional Hydrogen Bond (SER A:203, GLY A:122, TYR A:337, TYR A:124).
Polyvinyl chloride	−2.7	−2.7	4.210	**PyRx:** Alkyl (TRP A:439, TYR A:337, TYR A:449) and Pi-Alkyl(TRP A:86, PRO A:446). **CB-Dock:** Alkyl (PRO A:446) and Pi-Alkyl(TRY A:449, TYR A:337, TRP A:439). **eDock:** Pi-Alkyl(PHE A:338, HIS A:447, PHE A:295, PHE A:297).
Polycarbonate	−8.2	−8.5	6.143	**PyRx:** Conventional Hydrogen Bond (TYR A:124, TYR A:72, TYR A:337) and Pi-Donor and Hydrogen Bond (TYR A:124) and Pi-Pi Stacked(TRP A:286) and Pi-Pi T-shaped(TYR A:341). **CB-Dock:** Conventional Hydrogen Bond (GLY A:120, TYR A:133), Pi-Donor Hydrogen Bond (TYR A:337), Pi-Pi Stacked(TRP A:86), Pi-Alkyl(TYR A:124, HIS A:447). **eDock:** Pi-Donor Hydrogen Bond (TRP A:86) and Pi-Alkyl(TYR A:337).
Polyethylene terephthalate	−6.4	−7.4	5.673	**PyRx:** Conventional Hydrogen Bond (TYR A:72, TYR A:337, TYR A:124) and Pi-Pi Stacked(TRP A:286, TYR A:341). **CB-Dock:** Conventional Hydrogen Bond (SER A:203, TYR A:449, TRP A:439, TRP A:86, TYR A:337), Carbon Hydrogen Bond (GLU A:202), Pi-Sigma(TYR A:337) and Pi-Sigma(TRP A:86). **eDock:** Conventional Hydrogen Bond (TYR A:337) and Carbon Hydrogen Bond (TYR A:124, TYR A:337, SER A:203) and Pi-Pi T-shaped(TYR A:341).

Continued on next page

Table 2 (continued)

Plastic components	PyRx binding affinity (kcal/mol)	CB-Dock Vina score	eDock X-Score	Bonds and interactions
Polyamide (nylon–66)	−5.6	−6.9	5.203	**PyRx:** Conventional Hydrogen Bond (TYR A:72, GLN A:291, SER A:293) and Alkyl(TYR A:341, TYR A:124) and Pi-Alkyl(VAL A:294, TRP A:286). **CB-Dock:** Carbon Hydrogen Bond (TRP A:86), Conventional Hydrogen Bond (SER A:203, TYR A:337, TYR A:124), Pi-Sigma(TRP A:439), Pi-Alkyl(TYR A:449). **eDock:** Conventional Hydrogen Bond (TYR A:337, TYR A:124, ASN A:87) and Carbon Hydrogen Bond (TYR A:337) and Pi-Alkyl(HIS A:447, PHE A:338, PHE A:297, PHE A:295).

3.2.2 Molecular docking using CB-dock server

CB-Dock was used to study the binding modes and interactions of the seven common microplastics within the AChE active site. The docking scores present in Table 2 in kcal/mol revealed polycarbonate had the lowest, most favourable binding energy of -8.5 kcal/mol, followed by Dodecane and PET with scores around -5.8 and -7.4 kcal/mol respectively. In contrast, PVC was predicted to bind the weakest between -2.7 to -1.8 kcal/mol, over 3 kcal/mol less favourable than PC. This substantial energy gap implies PVC forms considerably less stable interactions within the gorge compared to the other plastics, possibly due to its inability to compensate entropically for fitting in the tight active site. Examining the interaction types provided insights into factors governing relative binding preferences. PC participated in an extensive hydrogen bonding network as well as pi-pi, pi-donor and pialkyl interactions, optimally engaging key catalytic site residues like Tyr124, Trp86 and His447 (Table 2). Such a combination of complementary interaction modes likely confers its most stable binding poses. Dodecane and PET also formed hydrogen bonds and pi-sigma stacking in addition to hydrophobic contacts, with their lower yet still favourable scores potentially derived from a balanced interacting profile fitting the gorge architecture. In contrast, PVC showed only simple alkyl and pialkyl interactions with limited residues including Trp439 and Tyr337/ 449, suggesting pi forces alone are insufficient to compensate entropy loss upon complexation, accounting for its poor estimated affinity. Overall, CB-Dock predicted PC, Dodecane and PET as the most stably binding microplastics based on their favourable scores and optimal hydrogen bonding, pi and hydrophobic contacts fully engaging the gorge, supporting experimental study into possible AChE inhibition and toxicity.

3.2.3 Molecular docking using eDock server

eDock was used to predict the binding modes and affinities of the seven common microplastics within the AChEactive site. Dodecane exhibited the highest X-SCORE of 5.310, indicating its most favourable predicted binding pose. Polycarbonate followed with a score of 6.143, while PU and PP scored comparably around 4.3–4.5. PVC had the lowest affinity of 4.210 based on eDock scores (Table 2). Examination of interaction types between each plastic and protein residues revealed dodecane and polycarbonate formed the most diverse interactions within the active site. Dodecane engaged in hydrophobic interactions with Val294 in addition to pi-alkyl stacking with His447, Phe338, Tyr124, Phe297, Tyr341 (Table 2). Polycarbonate showed a combination of pi-donor hydrogen bonding via Trp86 and pi-alkyl interactions with Tyr337. Such diverse interacting modalities likely contributed to their high docking scores. The other microplastics were found to primarily involve pi-alkyl stacking between aromatic residues in the gorge region, with polyurethane, PP and polyamide engaging His447, Phe338 and Tyr124/337, and PVC forming pi-alkyl contacts with Phe338, His447, Phe295 and Phe297 (Table 2). Pi interactions allow optimal orientation near the catalytic triad. Despite scoring lower, PVC and polyamide contained only pi-alkyl interactions qualitatively similar to higher scoring ligands, indicating subtle energetic or geometric factors beyond interacting functionalities account for affinity differences. Overall, eDock predicted dodecane and polycarbonate as the most stable binders based on interacting diversity, score values and visual analysis of poses, with their binding modes incorporating a balanced combination of hydrophobic, pi and hydrogen bonding interactions optimally engaging enzyme residues.

(continued on next page)

The results from multiple molecular docking algorithms predicting the binding of various microplastics to AChE provide interesting insights into how these emerging pollutants may interact with important enzyme targets. It is important to compare and validate the predictions from different docking programs to attain more reliable results (Wang et al., 2016). In this study, three molecular docking servers (PyRx, CB-Dock, and eDock) were utilized to model the interactions between seven microplastics (PC, PU, PVC, PP, PS, nylon, and dodecane) and human AChE. Every program has different docking algorithms as well as scoring functions therefore comparing combined predictions provides a more computed analysis than any individual method can do (Kitchen et al., 2004). The PyRx results identified polycarbonate as forming the most favourable interactions within the AChE active site, with an estimated binding affinity of -8.2 kcal/mol. Aligned with the PyRx results, CB-Dock and eDock also predicted high binding affinities for polycarbonate and Dodecane, identifying them as stable binders due to their ability to interact through multiple supportive binding modes within the binding site (Rana et al., 2017). Dodecane, a common component of PE microplastics debris, was also shown to be more toxic by admetSAR2 and AdmetLab. Across all three docking servers, PVC consistently showed the weakest predicted binding to AChE with a binding affinity of -2.7 kcal/mol. However, PVC has been shown to be more toxic than other plastics. Recent studies have further validated the key insights from this computational analysis. A 2024 publication by Melki et al., corroborated the high toxicity and AChE inhibitory potential of Bisphenol A microplastics through in vitro experiments with Cholinergic Function for the Ascidian Ciona robusta. They demonstrated Bisphenol A induced significant oxidative stress, and impairment of cholinergic signalling, aligning with the predicted strong binding to the AChE active site. In some other research, some plastic forms of PVC have impaired memory and development in fish indicating that it may degrade cholinergic function as well (Buzenchi Proca et al., 2024; Santoro et al., 2024).

3.3 Molecular interaction of substrate with AChE

The molecular docking of substrate HUPRINE W with the AChE receptor was performed using PyRx software. The results showed a high binding affinity of -8 kcal/mol between HUPRINE W and the receptor binding site. Upon examining the ligand interactions, it was observed that HUPRINE W formed five conventional hydrogen bonds with the receptor residues TYR A:124, TYR A:341, SER A:293, and SER A:293, GLN A:291. Additionally, there were three Pi-Alkyl bonds formed between HUPRINE W and the receptor residues TYR A:341 and TYR A:341, TRP A:286. Lastly, three Pi-Pi stacked interactions were seen between HUPRINE W and the receptor residues PHE A:297, PHE A:338 and TRP A:286 within the binding pocket. These complementary bonding interactions involving hydrogen bonds, Pi-Alkyl and Pi-Pi bonds indicated the tight binding and stable docked conformation of HUPRINE W within the AChE receptor cavity predicted by PyRx software. As a reference, the AChE substrate HUPRINE W was also docked into the enzyme's active site (Nachon et al., 2013; Singh and Kumar, 2024). The comparable binding poses and interactions of HUPRINE W with key binding residues validate it as a positive control for evaluating inhibition (Camps et al., 2000). However, surprisingly the docking scores obtained for dodecane and polycarbonate were lower than HUPRINE W (-8 kcal/mol). This suggests that though HUPRINE W binds strongly to AChE as expected from experimental studies, dodecane and polycarbonate make stable complexes within the active site.

3.4 Pharmacophore modelling and molecular interaction

3.4.1 Pharmacophore modelling and screening

Using the default shape filtering settings, specialized libraries containing over 35 million compounds (CMNPD, COCONUT, NPASS, NANPDB, ZINC natural products and drugs) were rapidly searched. Pharmit's integrated pharmacophore shape filtering initially identified 33 hits satisfying the maximum number of Dodecane pharmacophore anchors (Supplementary Figure. 1).

The top 30 hits from each database (total 150 compounds) were further filtered based on constraints including six hydrophobic rings, RMSD between anchors and poses ≤2 Å, molecular mass 100–300 Da, and ≤5 rotatable bonds. The 10 hits from each database with the lowest RMSD to Dodecane anchors were shortlisted,

providing a final set of 50 compounds for molecular docking. The structure-based drug design approach which helps to identify chemical features responsible for biological activity(Yang, 2010).

3.4.2 Molecular interaction of AChE receptor with screened compounds using PyRx software

The 50 screened compounds retrieved from pharmacophore modelling were subjected to molecular docking simulations against the 4BDT AChE receptor using PyRx software. The receptor was prepared and the active site grid defined based on the co-crystallized dodecane position. Each ligand underwent energy minimization and flexible blind docking employing the LGA for conformational searching. The resulting docked poses were analysed for their binding affinities and interaction profiles, with Triacontafluoropentadec-1-ene from the ZINC database exhibiting the strongest predicted binding with a score -9.6 kcal/mol, indicating its complex with AChE is thermodynamically most favourable amongst the screened hits (Table 3). Examining the predicted binding interactions provided insights into factors governing optimized docking, where Triacontafluoropentadec-1-ene (PubChem id: 88922911) established a diverse network of interactions anchoring it stably within the gorge by engaging in van der Waals contacts with aromatic residues Trp286 and Leu289 near the entrance and forming a carbon hydrogen bond with Val294 deeper in the active site (Fig. 1A). It participated in an extensive array of conventional hydrogen bonds donated to residues critical for hydrolysis including Tyr337, Tyr124, Phe295, Arg296, Ser293 and Tyr72 lining the cleft. A distinguishing feature was its oligofluoro substituent, which contributed halogen interactions with Tyr341, Asp74, Arg296 and Ser293 via fluorine atoms (Fig. 1B). Finally, its alkyl chain participated in π-alkyl stacking with Phe297 and Phe338 near the catalytic triad. Re-docking of these substitutes with AChE revealed that Triacontafluoropentadec-1-ene (PubChem CID 88922911) obtained the most favourable score of -9.6 kcal/mol. Formation of an

Table 3 *Binding affinities of top 50 screened plastic components docked against the AChE protein receptor by using the PyRx molecular docking tool. This table presents the docking scores (kcal/mol) of the 50 best-scoring pharmacophore screened plastic components predicted to bind effectively on the active site of the AChE protein receptor as determined by PyRx docking simulations. High negative score corresponds to stronger predicted binding affinity.*

Filtered compounds	CMNPD	COCONUT	NANPDB	NPASS	ZINC
1st	−8.0	−6.7	−8.6	−6.6	−9.6
2nd	−8.1	−4.7	−8.0	−6.2	−9.3
3rd	−8.3	−6.9	−8.7	−6.0	−5.1
4th	−8.1	−5.5	−8.4	−5.8	−8.8
5th	−8.6	−6.8	−5.2	−7.0	−8.4
6th	−8.3	−5.7	−5.0	−5.1	−8.7
7th	−9.2	−5.8	−8.5	−7.5	−8.8
8th	−7.3	−7.0	−8.7	−5.7	−9.4
9th	−7.4	−6.2	−5.4	−7.5	−5.2
10th	−9.0	−6.5	−7.0	−6.4	−8.6

Table 4 *Binding affinities of the top 50 screened plastic components docked against the AChE protein receptor using the PyRx molecular docking tool. High negative scores correspond to stronger predicted binding affinity.*

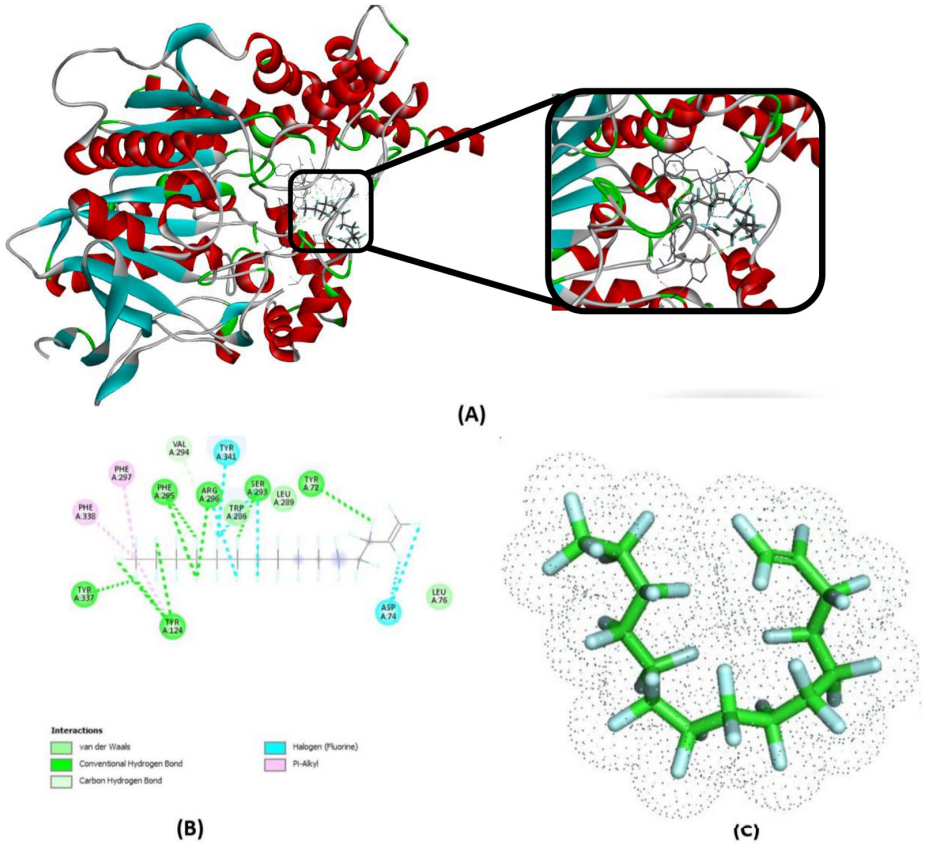


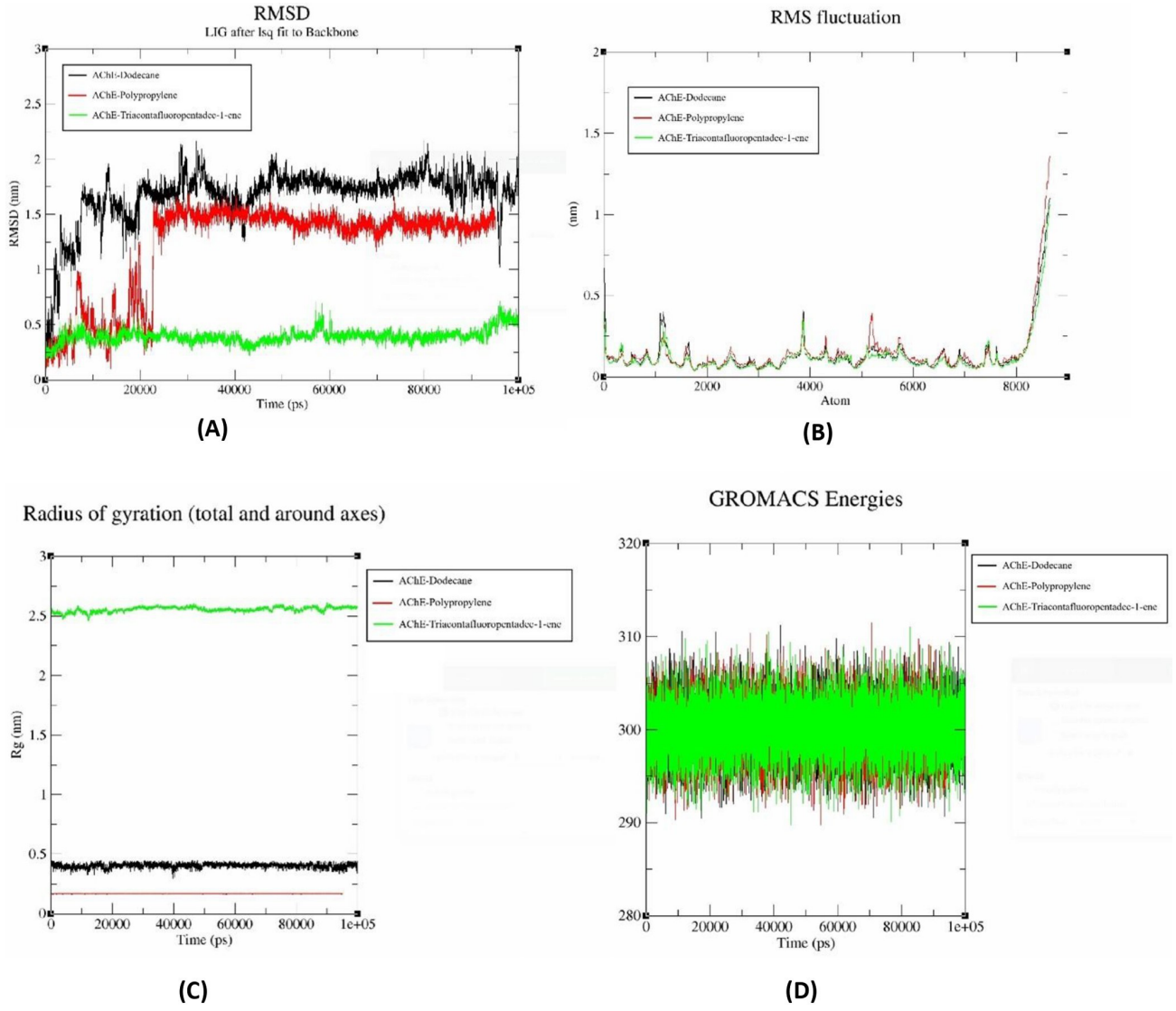
Figure 1 *(A) Detailed analysis of the AChE-Triacontafluoropentadec-1-ene docked complex showing the highest docking score of −9.6 kcal/mol and eight hydrogen bonds between protein (multi-coloured cartoon) and ligand (grey and blue sticks). The 3D protein-ligand interactions were visualized and analysed using Discovery Studio v24 software. (B) 2D ligand interaction diagram of the AChE-Triacontafluoropentadec-1-ene complex showing hydrogen bonds between the ligand (blue and grey sticks) and the protein visualized using Discovery Studio v24. (C) 3D structure of the Triacontafluoropentadec-1-ene ligand shown in sticks representation with carbon in green and oxygen in white.*

3.5 Molecular dynamics analysis of AChE docked complexes

The docking results from PyRx revealed dodecane interacting strongly within AChE (4BDT), forming favourable pi-stacked and alkyl contacts with key residues in the active site gorge including Tyr341, Trp286 and Tyr124. Of the different plastic components examined, dodecane exhibited the strongest predicted binding affinity of -5.1 kcal/mol and the PP exhibited -4.7 kcal/mol. These highest and lowest affinity poses were selected for extended atomistic molecular dynamics simulations using GROMACS in order to investigate the binding mechanism and stability over time at an atomic level of detail. The first simulation examined the AChE-dodecane complex over 100 ns. RMSD graph, showed in that initial fluctuations between 0.5 and 1.5 nm as hydrogen positions and sidechain orientations exhibited various conformations to minimize energy in the aqueous environment. Equilibration showing over 22 ns as visible by the decreasing RMSD values, peak in a stable plain between 1 and 1.5 nm (Fig. 2A). The radius of gyration distributed between 0.165 and 0.170 nm (Fig. 2C). This tight distribution demonstrated the van der Waals-dominated aliphatic plastic retained its rigid molecular structure without flexibility. Temperature was quantified by averaging total potential energy around -30,000 kJ/mol corresponding to 300 K, maintaining physiological conditions. The non-polar ligand preferentially associated through hydrophobic contacts rather than polar hydrogen bonding as verified by the empty H-bond graph. The second simulation of AChE-polypropylene analysed a more conformationally flexible plastic over 100 ns. Here, the initial RMSD values began lower between 0 and 0.5 nm since van der Waals forces loosely guided the linear polymer into an energetically favourable orientation. Equilibration was comparable albeit at a reduced value between 0.3 and 0.4 nm by 50 ns. Accounting for increased molecular size, radius of gyration measured 0.3–0.4 nm as expected. Temperature was stable at 295–305 K as energy fluctuated near -29,000 kJ/mol (Fig. 2D). A key distinction was Polypropylene's ability to mediate 1–2 hydrogen bonds per frame on average through polar groups, introducing directional stability beyond hydrophobic interactions afforded by Dodecane alone. This additional binding complement conceivably enhances residence time.

The top ranked hit from molecular docking, Triacontafluoropentadec-1-ene, was selected for all-atom 100 ns GROMACS molecular dynamics simulations of the AChE-Triacontafluoropentadec-1-ene complex. Trajectory analysis revealed key aspects of the binding mechanism over extended real time, with the RMSD of the ligand fluctuating between 0.2 and 0.4 nm, initially as interactions stabilized before settling in this range, indicating retention of the docked pose (Fig. 2A). Radius of gyration measured a consistent 0.54 nm, confirming the rigid ligand architecture. Total potential energy fluctuated around -29,500 kJ/mol corresponding to ambient 300 K, suggesting system thermodynamic stability. Examining heavy atom RMS fluctuations is pinpointed key interacting residues maintaining ≤0.2 nm values including Tyr341 and Trp286 near the catalytic triad engaged in π-alkyl stacking (Fig. 2C). The extended dynamics simulation visualized how the ligand established a balanced repertoire of complementary interactions optimally fitting enzyme architecture through a combination of directed H-bonding, halogen effects and dispersed hydrophobic contacts, conferring stable recognition over real dynamic timescales and providing atomic level rationale for its high affinity determined through docking(Naqvi et al., 2018). This supports Triacontafluoropentadec-1-ene merits further investigation as a potential AChE inhibitor.

Figure 2. Analysis of the AChE-Dodecane, AChE-Polypropylene and AChE-Triacontafluoropentadec-1-ene complexes from 100 ns molecular dynamics simulations: 2 (A) Backbone root mean square deviation (RMSD) analysis. The black curve represents the RMSD of the AChE-Dodecane complex, the red curve represents the RMSD of the AChE-Polypropylene complex, and the light green curve represents the RMSD of the AChE-Triacontafluoropentadec-1-ene complex. 2(B) Root mean square fluctuation (RMSF) analysis. The black curve represents the RMSF of the AChE-Dodecane complex, the red curve represents the RMSF of the AChE-Polypropylene complex, and the light green curve represents the RMSF of the AChE-Triacontafluoropentadec-1-ene complex. 2(C) Radius of gyration analysis. The black curve represents the radius of gyration for the AChE-Dodecane complex, the red curve represents the radius of gyration for the AChE-Polypropylene complex, and the light green curve represents the radius of gyration for the AChE-Triacontafluoropentadec-1-ene complex. 2(D) GROMACS total energy analysis.



The black curve represents the total energy for the AChE-Dodecane complex, the red curve represents the total energy for the AChE-Polypropylene complex, and the light green curve represents the total energy of the AChE-Triacontafluoropentadec-1-ene complex.

3.6 Molecular dynamics analysis of AChE docked complexes

Molecular dynamics simulations of the AChE-ligand complexes. Molecular dynamics simulations of the AChE-ligand complexes exposed, Dodecane-AChE relatively stable RMSD values after 22 ns of equilibration indicate it maintains the docked pose well through van der Waals interactions (Fig. 2A) (Pantsar and Poso, 2018). Polypropylene-AChE showed faster initial equilibration, its RMSD and radius of gyration values also stabilized, but slightly higher, reflecting its increased size. AChE-Triacontafluoropentadec-1-ene binding most stably within the active site residues through a balanced range of H-bonding, halogen, and hydrophobic interactions, maintaining an RMSD of 0.2–0.4 nm and interacting residues within 0.2 nm over the full 100 ns simulation at a total energy of -29,500 kJ/mol at 300 K. This optimized fitting in the enzyme through complementary interactions likely explains its high docking score. The MD simulations validated ligand poses and binding mechanisms over extended timescales, supporting the docking results (Naqvi et al., 2018). Flexible PP and rigid, well-optimized Triacontafluoropentadec-1-ene showed the most promising dynamic interactions and stability within AChE. This study provides a micromodel for further computational and experimental investigations into microplastics neurotoxicity. The combined in-silico approaches applied here could be expanded upon in future work. In addition, the top hits identified from pharmacophorebased screening require experimental validation of their predicted AChE inhibitory activity. Also, the current findings open opportunities to study microplastics interactions with other vital neuroreceptors beyond AChE. A key aim of this work was to identify a safer alternative substrate to dodecane for reducing harm from micro/nanoplastic synthesis in fisheries industries. Our data indicates that Triacontafluoropentadec-1-ene could be a replacement component of dodecane, necessitating further investigation into its feasibility for polymerization. This compound may potentially replace dodecaneincorporating products currently used in the fisheries industry. Furthermore, a deeper understanding of microplastics neurotoxicity and safer alternatives facilitated by this research can help inform policies maker to tackle the global plastic pollution problem from fisheries. It could potentially provide solutions by developing non-toxic and environmentally-benign plastic substitutes. Clarifying the relative neurotoxicity of different microplastics will also assist risk assessment. Overall, this work sets the stage for more targeted experimental studies with applications in regulations, remediation, and public health.

4. Conclusion

In this study, we computationally evaluated the potential toxicity mechanisms of seven commonly used microplastics in the fishing industry (nylon, PVC, polycarbonate (PC), polypropylene (PP), polyethylene terephthalate (PET), polyurethane (PU), and dodecane) through their interactions with the acetylcholinesterase (AChE) receptor. The findings revealed neurotoxic effects linked to these microplastics, with polycarbonate showing the strongest binding affinity of -8.2 kcal/mol, followed by dodecane at -5.1 kcal/mol and polypropylene of -4.7 kcal/mol. Both AChE-dodecane and AChEpolypropylene complexes established stability over 100 ns of MD simulations. Notably, triacontafluoropentadec-1-ene emerged as a promising “greener” alternative, achieving the highest binding score of -9.6 kcal/mol and exhibiting stability in molecular dynamics. These results highlight the revealed of triacontafluoropentadec-1-ene as a novel replacement for dodecane, AChE inhibitor and the framework provided new insights into the identification of hazardous microplastics components though accelerating the discovery of safer alternatives to enhance environmental safety. The findings need to be further validated through experimental evidence for the predicted interactions and toxicological effects. Furthermore, to fully understand the long-term environmental consequences and impact of triacontafluoropentadec-1-ene, is required in addition to its potential for large-scale production and adoption in the fishing industry. Future studies need to be explored about the potential interactive effects of the combined microplastics mixture and their potential impact on aquatic organisms and human health.

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Abbreviations

MP– Microplastics; NP – Nanoplastic; AChE - Acetylcholinesterase; ADMET - Absorption, Distribution, Metabolism, Excretion and Toxicity; PE - polyethylene; PP - polypropylene; PS - polystyrene; PVC - poly vinyl chloride; PU - polyurethane; PCBs - Polychlorinated biphenyls; DDT - Dichlorodiphenyltrichloroethane; PAHs - Polycyclic aromatic hydrocarbons; LGA: Lamarckian Genetic Algorithm; CMNPD - Chemical Modelling and Natural Products Database; NPASS - Natural Product Activity & Species Source Database; 2D- 2-Dimensional; 3D- 3-Dimensional; ADME- Absorption, Distribution, Metabolism, Excretion; GROMACS- Groningen Machine for Chemical Simulation; MD- Molecular Dynamics; CID- Compound Identification; PCBs-polychlorinated biphenyls; DDT-dichlorodiphenyltrichloroethane; PAHs-polycyclic aromatic hydrocarbons; SDF- Structure Data File; DSS- Decision Support System; CASTp - Computer Atlas of Surface Topography of Proteins; PyMOL- Python-enhanced Molecular Graphics; ERRAT- ERRors and Amino acid Types; PDB- Protein Data Bank; RMSD-Root Mean Square Deviation; RMSF-Root Mean Square Fluctuation; NIH- National Institute of Health; NCBI- National Centre for Biotechnology Information.

CRediT authorship contribution statement

Sindhuprava Rana: Writing – review & editing, Supervision, Software, Resources, Conceptualization. Ganesh Chandra Sahoo: Writing – review & editing, Supervision, Software, Resources, Project administration, Data curation, Conceptualization. Rajnarayan R Tiwari: Writing – review & editing, Supervision, Resources, Project administration, Formal analysis. Rahul Thakur: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. Vibhor Joshi: Writing – review & editing, Supervision, Resources, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.compbiolchem.2025.108358.

Data availability

Data will be made available on request.

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